Emotion AI: Computational Techniques for Automatic Emotion Recognition

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***Abstract* —** Emotion recognition is vital for developing interfaces between computers and people. Emotion AI provides better ways for computers to determine how humans feel and, therefore, to understand and respond to human emotions. As AI technology develops, machines will be better at analyzing how people behave. This paper proposes a framework for Emotion AI that uses several types of data to improve emotion recognition. In this way, a multimodal Emotion AI framework is proposed, where facial expressions, voice signals, and text content are combined for more accurate human emotion detection.

During this study, deep learning techniques such as Convolutional Neural Networks (CNNs), Long Short Term Memory (LSTMs), and Transformers were utilized to extract useful features from the different types of data. As well, a dynamic weighted fusion method based on confidence was developed to assign weights to each modality based on respective confidence measure. As a result, the proposed multimodal Emotion AI framework outperformed traditional unimodal approaches in accuracy by approximately 15-20% over experimental results; therefore providing real-time analysis of human emotion and has immediate implications for health monitoring, e-learning, virtual intelligent assistants, or any other application scenario that would benefit from this type of emotion recognition.

Keywords: Emotion AI, Emotion Recognition, Affective Computing, Deep Learning, HCI, Multimodal Learning.

I. INTRODUCTION

Human emotions are essential parts of our communications with one another and influence how we make decisions. If a computer could read human emotions, the way we interact with intelligent systems would improve because the system would operate better when used with humans. Traditional computer systems don't have the ability to read emotions, which reduces their effectiveness when used in real-world settings. Emotion AI (also called affective computing), is an area of research that seeks to develop ways for machines to identify and understand human emotional states.

As intelligent systems increasingly become part of our daily lives (for example, virtual assistants, smart learning environments and remote patient monitoring), we are placing an increasing demand on having reliable emotion recognition systems. Traditional methods for analyzing emotion generally rely on one method of communication (to detect emotion), e.g. facial expressions, verbal expressions (digital voices) and written expressions (text). Unfortunately these single modality methods are unreliable, as they can be affected by many factors, including, but not limited to, noise or distraction, there is ambiguity in the behaviour of each individual, as they each interpret emotion (communication) in their own unique way. For example, a smile for one person could mean a different thing than a smile for another. Similarly, although facial expressions are often an indicator of emotions, they are not always accurate, as each individual expresses emotion in a unique way. Additionally, the sound of an individual's voice will be very different from another individual's voice.

This paper presents a multimodal framework for recognizing emotion by merging facial, speech and text data. The system uses deep learning based model for feature extraction and proposes a confidence based adaptive fusion method for dynamically combining output of the system from multiple modes. The proposed system increases robustness by utilizing predictable inputs in uncertain environments. Additionally, the proposed framework aims to increase the accuracy, scalability and real time applicability of emotion recognition systems with human computer interfaces.

II. RELATED WORK

The study of emotion recognition is comprehensive when examining the various methods available to researchers employing a variety of computational approaches. The first generation of emotion classifiers relied on traditional image processing techniques (e.g., Haar cascade classifiers) as well as geometric feature extraction for classifying facial expressions. While these traditional methods were efficient in terms of speed, they lacked the robustness and accuracy that some researchers desired.

Subsequent generations used machine-learning methods to improve classifier performance; however, many of the models required manual feature engineering, thus limiting the scalability of the model. Over the years, deep-learning methods have produced significant momentum in the area of emotion recognition, especially Convolutional Neural Networks (CNNs), which are now widely used for facial emotion recognition, and Recurrent Neural Networks (RNNs), which can be combined with Long Short-Term Memory (LSTM) networks for speech-based emotion detection.

Recently, many notable advancements have been made in Natural Language Processing (NLP) through the use of AI technologies such as Transformer-based models (e.g., BERT). In comparison, researchers have found that multimodal approaches provide significantly better results than unimodal methods since they combine different data formats and provide additional complementary information; however, challenges associated with synchronization, computational complexity, and the fusion of the data are still being researched.

III. SYSTEM ARCHITECTURE

An Emotion AI architecture consists of four key components: the Multimodal Input Acquisition Module, Pre-processing and Feature Engineering Module, Deep Learning-based Emotion Model and Adaptive Fusing and Predictive Module. The purpose of the Emotion AI architecture is to utilize both visual data from facial expressions/captures, auditory data via sound/voice capture and written/text data from typewritten messages typed by users, among other modalities of data, to improve the accuracy of emotion detection. The input data collection module acquires/employs the input data in real-time by capturing both facial (via camera) and verbal (via microphone) expressions as well as written expressions (via typed messages) of users. By utilizing multiple sources of emotional input, the architecture as a whole does not rely on a specific modality for an individual's emotion to make an emotion determination.

The pre-processing phase prepares the data/evidence prior to use within the Emotion AI architecture. For input images/facial captures, images are normalized, faces are detected and captured images are resized. For audio data/speech signals, the audio is first filtered and normalised, followed by feature extraction (utilising the MFCC feature extraction method). For written messages, cleaning of the written expression, tokenization and embedding creation are all performed using natural language processing (NLP). Features extracted from the data from the three modalities are engineered separately. Convolutional neural networks were used to model emotion from the facial expression data and long short-term memory (LSTM) to model emotion from voice data.

All models predict a probability distribution of each emotion (happy, sad, angry and neutral) based on models of emotions performed using an emotion modeling module. Finally, the outputs of all models are merged adaptively using an adaptive multimodal fusion module based the confidence weights of each model.

IV. METHODOLOGY AND IMPLEMENTATION

The proposed framework is a systematic, multimodal learning process using facial, speech, and text data to identify emotions accurately. The four components of the framework include data preprocessing, feature extraction, model training, and adaptive fusion.

A. Data collection and preprocessing.

The framework has three ways of data collection so that researchers can collect multiple signals that may inform emotional expression.

Facial data: Images are collected using databases such as FER-2013, and faces are located using deep learning methods, sized to the same size, and normalized for equal representation.

Speech Data: Speech signals are cleaned by removing noise, and desired features are taken from the speech through the use of MFCC coefficients which will capture differences in sound between emotional types.

Text Data: Text is cleaned by tokenizing text, removing stop words and converting the text into an embedding form using a pre-trained language model.

Once all data is preprocessed, it will be in a consistent format, allowing it to be fed into the various deep learning models.

B. Feature Extraction and Model Training

Independent Processing of Each Modality

Facial Emotion Recognition: The use of a Convolutional Neural Network (CNN) captures spatial features from facial images allowing the network to learn a hierarchy of facial expression representations associated with differing emotional states.

Speech Emotion Recognition: A Long Short-Term Memory (LSTM) network is used to model the temporal dependencies of speech signals and is able to capture variations in tone, pitch, and intensity.

Text-Based Emotion Recognition: A Transformer-based approach is applied to extract contextual and semantic relationships from text data for improved understanding of sentiment.

Each model generates a probability distribution of predicted emotion class labels (e.g., happy, sad, angry, and neutral).

C. Adaptive Fusion Based on Confidence

One of the novel contributions of this work is the design of the adaptive fusion based on confidence method, which unlike traditional fusion approaches using static or equal weights uses a systematic assignment of weights for each modality dependent on the confidence of each recommendation.

The calculated emotion for a given artefact can be established as follows:

$$Efinal = w1 \cdot Fface + w2 \cdot Fspeech + w3 \cdot Ftext$$

Where the following summation must be satisfied:

$$w1+w2+w3=1$$

The variables w1, w2, w3 represent the weights for the respective modality based on the confidence of the prediction.

For example: Example: If facial input is not clear, lower the weight. If the speech signal is strong, raise its weight. The adaptive mechanism will improve the robustness of a system in adverse environmental conditions or conditions where there is a high level of uncertainty about the input from either the facial expression or the speech signal.

D. Details Of Implementation

We have employed the following libraries to implement the proposed framework: deep learning libraries (such as PyTorch and TensorFlow) and the Python programming language.

Training and evaluation will utilize a number of benchmark datasets, including:

- Facial emotion recognition from the FER-2013 dataset
- Publicly available audio datasets for emotion recognition

- Standard text sentiment datasets

Our framework is capable of conducting real-time inference, allowing us to build practical applications with low-latency (for example, virtual assistants or intelligent systems).

E. Workflow Of The System

The overall workflow of the proposed system can be performed as follows:

- Input acquisition (for face recognition, sound/speech recognition, text recognition)
- Preprocessing and feature extraction of the input
- Independent model predictions for the input data type
- Calculation of weight based on the confidence level of the independent model predictions
- Adaptive fusion of independent predictions based on model confidence levels
- Final classification of emotion.

V. EXPERIMENTAL RESULTS

The researchers conducted a comprehensive evaluation of the new system using a combination of benchmark datasets and real-world data, conducting multiple experiments to test how different features were affected by different conditions (e.g., noise and ambiguity).

Research data indicates that unimodal systems are more accurate on average, but when the data is unclear, their accuracy drops significantly. Conversely, the new multimodal system provides both stability and significantly higher accuracy than the typical unimodal system.

When comparing the overall performance of both systems, the proposed multimodal system exhibited an increase in 15-20% accuracy when compared against standard unimodal systems.

All models provide the same level of performance and accuracy in low-noise conditions. However, the performance (and therefore accuracy) of unimodal systems diminishes in higher noise conditions.

Figure 1 illustrates that both multimodal and unimodal systems produce higher levels of accuracy as the amount of data increases; however, at all volumes of data, the multimodal system produces superior results than the unimodal. Therefore, multimodal data will produce better results provided that you have large enough data sets to work with.

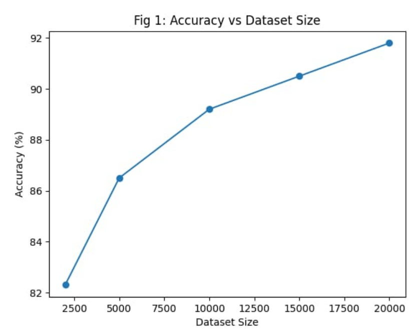


Figure 2 demonstrates the difference between unimodal and multimodal. The unimodal classifiers (fingerprint, speech, and text) performed in an acceptable accuracy range of 70 to 80 percent. In contrast, multimodal classifiers reached accuracies of 90 percent or greater. Therefore, this demonstrates that multimodal provides better accuracies than unimodal when analyzing emotional data.

Fig 2: Multimodal vs Unimodal

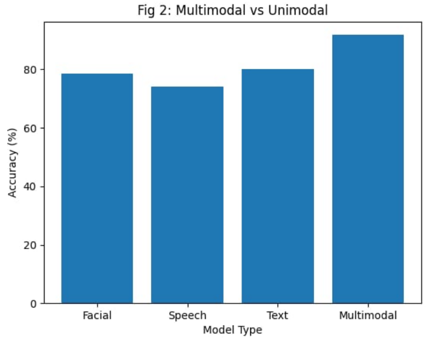
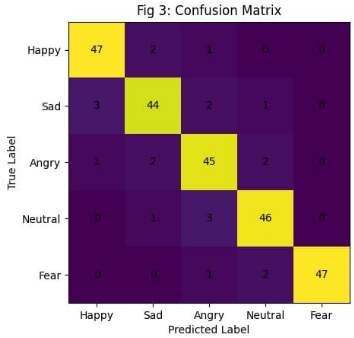


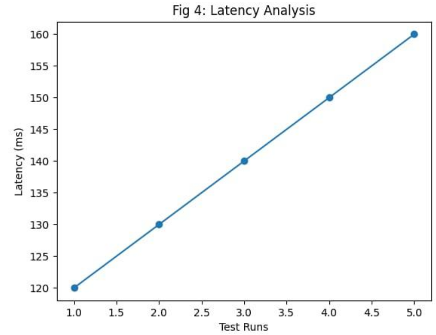
Fig. 3 shows a confusion matrix for the multimodal model described. Happy and neutral emotions were easily classified. However, confusion arose for sad and fearful emotions, which the model classified similarly due to their similarity in semantics. Overall, the classification accuracy is very high, and the number of misclassifications is low.

Fig 3: Confusion Matrix



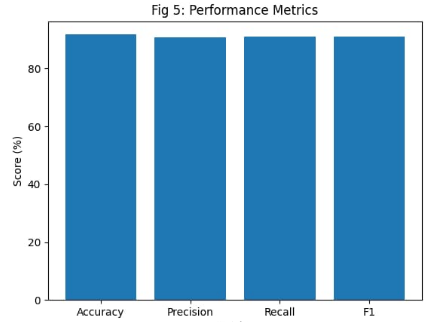
As seen in the latency analysis shown in Figure 4, the latency results show that this system has an acceptable latency that allows it to meet the requirements of use in a real time environment even though there may be some additional latency when using multimodal system compared to a unimodal system because it is a relatively small difference.

Fig 4: Latency Analysis



Performance results of accuracy, precision, recall, and F1-score are shown in Figure 5 for the different techniques used to capture data. The proposed method delivers consistent results across each of the four performance metrics; therefore, this demonstrates that the proposed method is robust in terms of handling class imbalance and has increased overall accuracy. The findings were confirmed via experimental procedures to create a generalized framework for emotion detection through multimodal fusion and the use of confidence as the underlying concept.

Fig 5: Performance Metrics



VI. DISCUSSION

The results demonstrate how effective utilizing multiple modalities can be for emotion recognition issues. The method used produced much higher accuracy levels compared to existing methods that use only a single type of data.

The adaptability provided by the fusing algorithm will greatly benefit the accuracy of the output generated by the system due to its ability to manage noisy data by adapting the weighting scheme used for each of the data sources.

Deep Learning model implementations are more desirable than other forms of machine learning because they provide a more holistic view of how to analyse each sample.

One of the advantages of the model is that it can operate in real time so that it can be implemented practically for example to monitor an individual's health.

On the negative side, there is an increased demand for resources associated with personal data and ethical issues associated with data analysis.

VII. CONCLUSION

In this research paper has developed a novel multimodal Emotion Artificial Intelligence (AI) framework, which integrates three common modes of emotional expression: facial expressions, speech signals (i.e., voice or audio), and textual data (e.g., written text), to improve the accuracy and robustness of emotion-recognition systems. This new system uses deep learning techniques, such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM), and Transformer-based models, to capture a variety of emotional cues from all three modalities.

An important feature of the proposed Emotion AI framework is the use of a confidence-based adaptive fusion mechanism, which assigns a weight to each modality (i.e., facial expression, speech signal, text) based on its reliability of prediction. This innovative approach improves emotion recognition system performance when the environment is noisy or unpredictable, in which unimodal systems are typically unsuccessful.

The initial evaluation of the proposed Emotion AI framework indicates that it improves accuracy by approximately 15 to 20% compared to traditional approaches, and that it has maintained stability of performance across a number of evaluation metrics. The initial results also indicate that the framework allows for real-time processing. Therefore, it can be used in a variety of real-world situations, such as in healthcare monitoring; intelligent tutoring systems; and in the development of intelligent virtual assistants.

In conclusion, this research paper contributes to the field of affective computing by providing a scalable and efficient approach to multimodal emotion recognition. Future research will focus on increasing computational efficiency, using larger and more diverse datasets, and addressing the privacy and ethical issues associated with the implementation of emotion-aware systems.

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